They stopped asking "what is a closure."

They started asking you to write this.

36 write-this questions

to

0 definitions

Uber · Swiggy · PayPal · Adobe · HackerRank · Atlassian

If Google can answer it in 8 seconds, it is out.

Sources: public Medium experiences, LinkedIn posts, FrontendGeek company loops.

CUT

What is hoisting

Virtual DOM, as a definition

let vs const

Controlled vs uncontrolled

Explain useMemo

What is a closure

KEPT

Implement it live

Debug without console

Name the follow-up

Design the product

Trade off out loud

Handle 10k rows

JAVASCRIPT

Functions they made people write.

01 **mapAsyncLimit**

Parallel. Order kept. Cap in-flight work. Uber

02 **Async memoize**

JSON key order + in-flight request dedupe. Uber

03 **Flatten a nested object**

Not an array. Adobe asked for the object.

04 **Flatten an array two ways**

Recursion and a stack. Complexity. Goibibo

05 **Infinite curry**

sum(1)(2)(3)() and mul(2)(3)(4)

06 **Promise.all polyfill**

Result order still has to be right.

A passing test is not the round.

07 **Event-loop OUTPUT**

Mix Promise, queueMicrotask, setTimeout. No definition.

08 **bind polyfill**

They still ask it. Write it once, cold.

09 **Deep equality for memo keys**

Same keys, different order, same cache hit.

10 **this + anonymous functions**

No console. One shot. Adobe

THE TRAP

They watch result order, failure mode, and whether a second identical call still hits the network.

Build these. Then survive the follow-up.

01 **Autocomplete**

Debounce, AbortController, stale response, keyboard, combobox ARIA.

02 **Nested checkbox tree**

Parents reflect children. Indeterminate state. Atlassian

03 **Nested comments**

Flat normalized state. Not a nested JSON blob. Flipkart

04 **OTP input**

Paste has to split across boxes.

05 **Pagination with ellipsis**

Active page, disabled edges. Razorpay

06 **Infinite scroll + 10k rows**

IntersectionObserver, then virtualize.

The ones that look done, and are not.

07 **File explorer**

Recursive expand. Lazy-load children.

08 **Kanban board**

HTML5 drag. Persist the board.

09 **Cart with live totals**

Quantity sync, not just a pretty list. PayPal / Swiggy

10 **Currency converter**

Dropdown + live convert. PayPal

11 **Multi-step review form**

Per-step validation, localStorage, toast. Flipkart

12 **Debug a broken sort modal**

Wrong comparator. No console. HackerRank

Green UI is the starting line.

If you cannot name the follow-up, you are not done.

Older fetch overwrites newer

Abort or ignore stale responses.

Unmount during a request

Cleanup is the round, not the happy path.

Keyboard, not just click

Arrow, Enter, Escape, focus stay on input.

Empty / loading / error / none

Four states. Most people ship one.

10k rows without jank

Window the list. Do not map 10k nodes.

Paste, not only type

OTP and chips die here.

Skip useToggle. Write these.

`useDebounce`

Cleanup the timer or you leak calls.

`useFetch`

Abort on unmount and on url change.

`useIntersectionObserver`

Infinite scroll, lazy images, sentinels.

`useOnClickOutside`

Modal, dropdown, context menu.

`usePolling`

Pause when the tab is hidden.

`useInfiniteScroll`

Observer + page cursor + error.

`usePrevious`

Compare renders without guessing.

`useLocalStorage`

Lazy init. JSON parse guard.

Not "explain useMemo."

01 **API fired twice**

Strict Mode vs a real bug. Prove which.

02 **When not to use useEffect**

Derived state, events, and data you already have.

03 **startTransition on the input**

Typing lags because you made the keystroke non-urgent.

04 **useDeferredValue vs useTransition**

Value lag vs marking an update interruptible.

05 **useLayoutEffect vs thrash**

Measure before paint. Do not use it as a faster effect.

06 **Compiler, RSC, diffing**

What the compiler skips. RSC vs SSR. PayPal asked for diffing pseudocode.

SYSTEM DESIGN

Walk in with a prompt, not a box fetish.

01 Swiggy restaurant page + live orders

02 WhatsApp Web - protocol, cache, a11y

03 Stock ticker - WS vs SSE vs poll

04 Google Docs - OT vs CRDT, offline

05 Typeahead as a product, not a demo

06 Data grid, 10k rows, live price

07 File system, 10k nodes

08 Seat map / BookMyShow

09 YouTube player + comments

10 Payment gateway states PayPal

SYSTEM DESIGN

They are not grading your boxes.

FETCH + CACHE

Where data lives. What invalidates it. What is stale.

REAL-TIME CHOICE

Poll, SSE, or WebSocket - and why this product.

VIRTUALIZE VS PAGINATE

10k nodes is a rendering problem, not a CSS problem.

EMPTY, RETRY, OFFLINE

The loop is won in the ugly states.

SPLIT THE ROUTE

Code-split the page, not every atom.

A11Y IS REQUIRED

Keyboard and ARIA are not a bonus slide.

NIGHT BEFORE

Write these eight. Then sleep.

01 mapAsyncLimit

02 Autocomplete with abort

03 Nested checkbox tree

04 useFetch with abort

05 Event-loop output

06 startTransition on the list, not the input

07 Swiggy restaurant + live orders (talk)

08 WhatsApp Web (talk)

Not a FAQ. A write list. Send it to whoever is interviewing tomorrow.